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Inventor(s)	Havlovitz; Paul M. et al.

INSECT TRAP

Abstract

An insect trap (**1**) may include a first housing (**10**); a reservoir (**40**) at least partially defined by the first housing (**10**); a protrusion disposed within the reservoir (**40**) and extending inward from an interior surface of the first housing (**10**), the protrusion (**42**) configured to pierce a package disposed within the reservoir (**40**); an actuator (**44**) connected to the protrusion (**42**) such that when the actuator (**44**) is moved, the protrusion (**42**) moves with the actuator (**44**) within the reservoir (**40**); a second housing (**60**) moveably engaging the first housing (**10**); a trap chamber (**70**) partially formed by the first and second housings (**10, 60**); an inlet (**72**) into the trap chamber (**70**); and wherein the second housing (**60**) is movable between a first position where the inlet (**72**) is open and a second position where the inlet (**72**) is closed. The trap (**1**) may include a package (**152**) enclosing an attractant composition. The package (**152**) has a release rate of the attractant composition from about 5 μg per day to about 500 μg per day.

Inventors: **Havlovitz; Paul M. (Dublin, OH), Ramoutar; Darryl (Delaware, OH), Ceddia; Michael Anthony (Dublin, OH), Rawlings; Matthew R. (Marysville, OH), Gries; Gerhard J. (Coquitlam, CA), Gries; Regine M. (Coquitlam, CA), LaFontaine; Jean Pierre (Delta, CA), Borden; John Harvey (Burnaby, CA), Britton; Robert A. (North Vancouver, CA), Crawley; Sydney E. (Dublin, OH)**

Applicant: **Scotts Canada Ltd.** (Toronto, CA)

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS [0001] This application claims the benefit of U.S. Non-provisional patent application Ser. No. 16/626,609, filed Dec. 26, 2019, which is a National Stage Entry Application Under 35 U.S.C. § 371 that claims the benefit of International Patent Application No. PCT/IB2018/000844, filed Jun. 29, 2018, which claims the benefit of U.S. Provisional Patent Application No. 62/527,528, filed on Jun. 30, 2017, the disclosures of which are hereby incorporated herein by reference in their entireties.

BACKGROUND
[0002] For many years, from the end of World War II until the final decade of the 20th century, much of the world was free from infestation by the common bed bug, *Cimex lectularius* (Heteroptera: Cimicidae). This is attributed to modern hygienic measures and the widespread use of DDT and other persistent insecticides. However, in the last two decades there has been a world-wide resurgence, with bed bugs becoming common urban pests, and sometimes causing debilitating skin irritation and lesions. This resurgence has renewed interest in detecting, monitoring, and controlling bed bug infestations.
SUMMARY
[0003] An insect trap may include a first housing; a reservoir at least partially defined by the first housing; a protrusion disposed within the reservoir and extending inward from an interior surface of the first housing, the protrusion configured to pierce a package disposed within the reservoir; an actuator connected to the protrusion such that when the actuator is moved, the protrusion moves with the actuator within the reservoir; a second housing moveably engaging the first housing; a trap chamber partially formed by the first and second housings; an inlet into the trap chamber; and wherein the second housing is movable between a first position where the inlet is open and a second position where the inlet is closed.
[0004] A kit for an insect trap includes a first housing having a first surface, a reservoir disposed within the first housing, a protrusion disposed within the reservoir and extending inward from an interior surface of the first housing, and an actuator disposed upon an exterior surface of the first housing and connected to the protrusion such that when the actuator is moved, the protrusion moves within the reservoir; a second housing moveably engaging the first housing; an attractant configured to attract insects and to be disposed within the reservoir; a substrate comprising an adhesive disposed upon at least a portion of a surface of the substrate, the substrate configured to adhere to the first surface of the first housing; a trap chamber formed by the first housing, second housing, and substrate when the substrate is adhered to the first surface of the first housing; and an inlet providing a passage into the trap chamber; wherein the second housing is movable between a first position where the inlet is open and a second position where the inlet is closed.
[0005] An insect trap includes a housing; a reservoir at least partially defined by the housing; a protrusion extending from an interior surface of the housing; an actuator disposed upon the housing

and configured to move the protrusion within the reservoir; a trap chamber at least partially formed by the housings and constructed to capture insects; an outer package disposed within the reservoir and impermeable to gas; an inner package enclosed within the outer package, the inner package being permeable to gas; and an attractant composition disposed within the inner package; wherein the inner package is configured to release the attractant composition at a release rate from about 15 μg per day to about 400 μg per day.
[0006] An insect trap includes a first housing; a second housing moveably engaged to the first housing; a trap chamber at least partially formed by the first and second housings and constructed to capture insects; an inlet providing a passage into the trap chamber; histamine disposed along or within the housing; and wherein the second housing is movable between a first position where the inlet is open and a second position where the inlet is closed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] The above-mentioned and other features and advantages of the present disclosure, and the manner of attaining them, will become more apparent and the disclosure itself will be better understood by reference to the following description of non-limiting embodiments of the disclosure taken in conjunction with the accompanying drawings, wherein:
[0008] FIG. **1** is an isometric view of an embodiment of an insect trap according to one or more embodiments, wherein a second housing is in a first positon;
[0009] FIG. **2** is an exploded view of the trap in FIG. **1**;
[0010] FIG. **3** is a top plan view of the trap of FIG. **1**, wherein the second housing has been removed for illustration purposes;
[0011] FIG. **4** is a front elevational view of the trap of FIG. **1**;
[0012] FIG. **5** is a back elevational view of the trap of FIG. **1**;
[0013] FIG. **6** is a left side elevational view of the trap of FIG. **1**
[0014] FIG. **7** is a right side elevational view of the trap of FIG. **1**;
[0015] FIG. **8** is a top plan view of the trap of FIG. **1**;
[0016] FIG. **9** is a bottom plan view of the trap of FIG. **1**, wherein a substrate has been removed from a bottom wall of a first housing;
[0017] FIG. **10** is a cross sectional side view of the trap of FIG. **1** taken at lines **10**-**10**;
[0018] FIG. **11** is a cross sectional side view of the trap of FIG. **1** taken at lines **11-11**;
[0019] FIG. **12** is a cross sectional side view of the trap of FIG. **1** taken at lines **12-12**;
[0020] FIG. **13** is a cross sectional side view of the trap of FIG. **1** taken at lines **13-13**;
[0021] FIG. **14** is an isometric view of an embodiment of the trap of FIG. **1**, wherein the second housing is in a second positon;
[0022] FIG. **15** is a cross sectional side view of another embodiment of the trap of FIG. **1**;
[0023] FIG. **16** is an exploded view of an insect trap according to one or more embodiments;
[0024] FIG. **17** is an exploded view of an insect trap according to one or more embodiments;
[0025] FIG. **18** is a cross sectional side view of the trap of FIG. **17** taken at lines **18-18**; and
[0026] FIG. **19** illustrates the design of (a) the three-dish, dual-choice olfactometer, and (b) the large Plexiglass arena; the inserts illustrate the bed bug shelter consisting of histamine-impregnated or control filter paper (FP), corrugated cardboard (CC), and an inverted vial lid (IVL) containing volatile pheromone components formulated in mineral oil or mineral oil alone.
DETAILED DESCRIPTION
[0027] One or more embodiments shown and described herein provide a trap for capturing and/or trapping insects (e.g., bedbugs), and in particular, traps used in conjunction with effective compounds, compositions and methods for attracting, capturing, and/or arresting bed bugs.

Generally, one or more embodiments of an insect trap shown and described herein may include a housing, a reservoir at least partially defined by the housing and configured to hold one or more insect attractants, a protrusion within the reservoir, an actuator configured to actuate the protrusion, a trap chamber at least partially formed by the housing and comprising a floor, and an inlet into the trap chamber. In some embodiments, the trap chamber may include a “pit” (as will be explained below) that the insects entering the trap chamber via the inlet may fall into and are then unable to escape. In some embodiments, the trap chamber may include an adhesive on the floor or ceiling of the trap chamber trapping insects entering the trap chamber via the inlet. In some embodiments, the trap chamber may include some combination of both a “pit” and adhesive to trap the insects. In some embodiments, the trap may include one or more attractants as will be described detail below herein. In one or more embodiments, the housing may be a unitary, monolithic structure or include multiple components making up the housing. Additionally, in one or more of the embodiments, the housing and floor may be a unitary, monolithic structure or include multiple components that are fixedly or removably connectable to each other.

[0028] Referring to FIGS. **1-14**, an embodiment of an insect trap **1** is shown. The insect trap **1** generally may include a first housing **10**, a reservoir **40** at least partially disposed within or defined by the first housing, a protrusion **42** disposed within the reservoir, an actuator **44** connected to and configured to actuate the protrusion, a second housing **60** moveably engaged to the first housing **10**, a trap chamber **70** partially formed by the first housing and second housing, and an inlet **72** providing an entrance into the trap chamber. The trap chamber may include a floor **100**. In some embodiments, the second housing **60** is configured to move relative to the first housing **10** between at least two positions: a first position (e.g., an open position) and a second position (e.g., a closed position). In the first position, the inlet **72** is open, permitting insects to enter and/or exit the trap chamber **70** shown in FIGS. **1**, **4**, **5**, **10**, and **11**. In the second position, the inlet **72** is closed, preventing insects from entering and/or exiting the trap chamber **70** shown in, for example, FIG. **14**. The trap **1** may include a longitudinal axis (L-L′) as shown.

[0029] FIG. **3** shows the first housing **10** with the second housing **60** removed. The first housing **10** may include a first portion **12**, a second portion **14** spaced apart from the first portion, a bottom wall **16** connecting the first and second portions **12**, **14**, a space **18** disposed between the first and second portions **12**, **14**, and a longitudinal chamber member **22** also connecting the first and second portions **12**, **14**. In some embodiments, the first portion **12** may include a first vertical wall **34***a*, second vertical wall **34***b* connected to the first vertical wall, a third vertical wall **34***c* connected to the second vertical wall, and a fourth vertical wall **34***d* connected between the first and third vertical walls. All four vertical walls **34***a-d* may extend upward from the bottom wall **16** to at least partially form or define the reservoir **40** disposed therein.

[0030] The first portion **12** may also include an upper housing **30** that that is configured to slide over the top of, and at least partially enclose, the four vertical walls **34***a-d* therein. In such a configuration, the reservoir **40** is completely disposed within or enclosed by the first portion **12** of the first housing **10** and the upper housing **30**. The reservoir **40** may be configured to receive and hold an attractant **150** therein. As will be described in greater detail below herein, in some embodiments, the attractant **150** may include an attractant composition, a first package **152** enclosing the attractant composition, and a second package **154** enclosing the first package **152**.

[0031] The upper housing **30** may include a first upper housing wall **38***a*, a second upper housing wall **38***b* connected to the first upper housing wall, a third upper housing wall **38***c* connected to the second upper housing wall, and a fourth upper housing wall **38***d* connected between the first and third upper housing walls. The upper housing **30** further includes the actuator **44** that forms a top wall of the upper housing that may be connected to upper ends of the four upper housing walls **38***a-d*. In some embodiments, when the upper housing **30** is placed over and slid down over the first portion **12** of the first housing **10**, the four vertical walls **34***a-d* of the first housing **10** are disposed and enclosed within the respective four upper housing walls **38***a-d* as shown, for example, in FIGS. **2** and **10-13**. The third upper housing wall **38***c* may include one or more upper housing tabs **20** extending outward from the third upper housing wall **38***c* toward the space **18**. The four vertical walls **34***a-d* and the four upper housing walls **38***a-d* may have any length and height.

[0032] In some embodiments, the upper housing **30** may be removably or fixedly connected to the first portion **12** of the first housing **10**. Illustrative connections may include adhesive, bolt/nut connections, screw connections, rivet connections, welds, tongue and groove connections, snap-fit connections such as, for example, tab and detent connections, and other similar type connections. As shown in FIGS. **2**, **12**, and **13**, in one embodiment, first housing **10** may include detents **11** disposed on second vertical wall **34***b* and fourth vertical wall **34***d*, and the upper housing **30** may include tabs **13** extending inward from second upper housing wall **38***b* and fourth upper housing wall **38***d*, which are configured to engage the respective detents **11** in the first housing **10**, connecting the upper housing **30** to the first housing **10**.

[0033] In some embodiments, the actuator **44** may include a button **46**, a protrusion holder **48** disposed on an underside of the button **46** and configured to hold and/or connect to the protrusion **42** such that the protrusion extends inward into the reservoir **40**. In some embodiments, the protrusion **42** may be configured to pierce a package such as, for example, one or more package walls, substrates, and/or layers of the attractant **150**. As shown, for example, the protrusion **42** is an elongated member (e.g., a rod) extending from the button **46**. The protrusion **42** may comprise a variety of shapes, materials, and configurations, including metal and/or plastic. The protrusion **42** may be separately connectable to the protrusion holder **48** or be fabricated from such that the protrusion **42** has a unitary, monolithic construction with the button **46** and/or the protrusion holder **48**. As such, in some embodiments, the protrusion **42** may be fabricated from the same material (e.g., polypropylene) as the button **46** and/or the protrusion holder **48**. In some embodiments, the protrusion **42** may be a pin. In some embodiments, the protrusion **42** may be configured to include a tip that is pointed or sharp.

[0034] The actuator **44** may also include a guard **50** that extends upward away from the first housing **10**. The guard **50** may surround the button **46** and include an opening **52** configured to permit a user to push on the button **46** through the opening **52** to actuate the button **46**. In some embodiments the guard **50** may be configured to protect the button **46** from accidental activation.

[0035] The actuator **44** and/or button **46** may be fabricated from a material that is resilient such that when a user applies a force to (e.g., pushes) the button **46** inward toward the reservoir **40**, the button moves, bends, or flexes inward toward the reservoir. In some embodiments, actuator **44** and/or button **46** may be fabricated from a material that has memory such that when a user applies a force to (e.g., pushes) the button **46** inward toward the reservoir **40**, the button moves, bends, or flexes inward toward the reservoir, causing the protrusion **42** to move linearly inward along a vertical axis (e.g., a downward direction) as illustrated by arrow A, ultimately piercing and/or puncturing a package (e.g., second package **154** of attractant **150**) disposed within the reservoir **40**. In some embodiments, the button **46** has a convex curvature or arch. Thus, when the button **46** is pushed inward toward the reservoir **40**, the button flexes inward or, in other words, the convex curvature collapses inward toward the reservoir **40** as described. Once the force is removed from the button **46**, the button **46** may return to its original shape and/or position (e.g., normal position) due to the material's memory. This movement back to the button's normal positon may also move the protrusion **42** back upward along the vertical axis out of and/or clear of any package disposed within the reservoir **40** (e.g., second package **154** of the attractant **150**). For example, the protrusion **42** is moved out of the puncture hole created by the protrusion in the second package **154** held within the reservoir **40**. To assist with the flexing and/or bending, in some embodiments, the actuator **44** and/or button **46** may include at least one area of flexibility or weakened areas **45**. In some embodiments, the weakened areas **45** may act as vents. Additionally, in some embodiments, weakened areas **45′** may be disposed in a wall of the first housing **10**, such as, for example, the third vertical wall **34***c* as shown in FIG. **16**.

[0036] The second portion **14** of the first housing **10** may include a second portion wall **15**. The second portion wall **15** may include one or more second portion tabs **17** extending outward therefrom toward the space **18**. The space **18** may be defined between the third upper housing wall **38***c* of the upper housing **30** and the second portion wall **15** of the second portion **14**.

[0037] As shown in FIGS. **2**, **9**, and **11** the longitudinal chamber member **22** may be disposed in the trap chamber **70** (and/or the space **18**) and extend upward into the trap chamber to form a wall disposed within the chamber along the longitudinal axis L-L′. The longitudinal chamber member **22** may have any length and height. Also, a plurality of interior channel walls **24** extend perpendicularly from the longitudinal chamber member **22** to form a plurality of channels **26** disposed between each set of adjacent interior channel walls, between the third upper housing wall **38***c* and one of the plurality of interior channel walls **24**, and between the second portion wall **15** and one of the plurality of interior channel walls **24**. The plurality of interior channel walls **24** may have any length and height. The plurality of channels **26** forms a plurality of sub-chambers within the trap chamber **70**. Referring to FIG. **3** (a top view of the trap **1** with the second housing **60** removed for illustration purposes only), the bottom wall **16** may include a trap chamber aperture **21** disposed there through and sized to match or substantially match the length and width dimensions of the trap chamber **70**. The longitudinal chamber member **22** and the plurality of interior chamber walls **24** divide the trap chamber aperture **21** into a plurality of sub-chamber apertures **23** that vertically align with each of the sub-chambers or channels **26** as best shown in FIGS. **3**, **9**, **10**, and **11**.

[0038] The first housing **10**, and one or more of its components, may be fabricated as a unitary, monolithic component. In some embodiments, the first housing, and one or more of its components, may be fabricated as multiple components that may be connected (e.g., fixedly or removably) to each other, using any connection method described herein and/or any other conventional or yet-to-be developed connections and/or connection techniques. As shown in the figures, all of the first housing **10** may be fabricated as a unitary, monolithic component, except for the upper housing **30** of the first housing **10**, which may be connected to the first portion **12** of the first housing as set forth above herein. In some embodiments, the first housing **10** is molded from plastics such as, for example, polypropylene, polyethylene, elastomer, copolymer, impact modified copolymer, polystyrene, combinations thereof, or the like, using conventional plastic molding techniques such as, for example, injection molding, blow molding, compression molding, extrusion molding, laminating, reaction injection molding, matrix molding, rotational molding (or rotomolding), 3D printing, combinations thereof, or the like. The first housing **10** and/or its features and components may comprise any number of configurations, sizes, shapes, components, designs, and/or other materials (e.g., composites, metals, etc.).

[0039] The second housing **60** may include a first wall **62**, a second wall **64** opposite the first wall, and a bottom edge **61** along both the first and second walls **62**, **64**. The first wall **62** may include one or more detents (e.g., first wall detents **66** disposed within an outer surface of the first wall **62** as shown in FIGS. **10** and **11**). Similarly, the second wall **64** may include one or more detents (e.g., second wall detents **68** disposed within an outer surface of the second wall **64** as shown in FIGS. **2**, **10**, and **11**). As shown, the second housing **60** may be inserted into the space **18** between the first portion **12** and second portion **14** of the first housing **10** such that upper housing tabs **20** engage respective first wall detents **66** on the first wall **62** of the second housing **60** and second portion tabs **17** engage respective second wall detents **68** on the second wall **64** of the second housing **60**. Such engagement of the tabs **20**, **17** of the first housing **10** with the respective detents **66**, **68** of the second housing may hold the second housing in the first position (as shown in FIGS. **1**, **4**, **5**, **10**, and **11**) connected to the first housing **10** within the space **18** as set forth above herein, forming and/or defining at least a portion of the trap chamber **70** (shown in FIGS. **10** and **11**). The inlet **72** is defined between the bottom edge **61** of the second housing **60** and the bottom wall **16** of the first housing **10** as shown in FIGS. **1**, **4**, **5**, and **10**. In some embodiments, a portion of the wall **16** is positioned above the inlet **72**. For example, as shown in FIG. **16**, a portion of the wall **16**′ is at a higher vertical elevation than the inlet **72** into the trap chamber. Thus, the inlet **72** may be defined between a bottom edge of the wall **16**′ and the floor **100**.

[0040] The second housing **60**, and one or more of its components, may be fabricated as a unitary, monolithic component. In some embodiments, the second housing, and one or more of its components, may be fabricated as multiple components that may be connected (e.g., fixedly or removably) to each other, using any connection method described herein and/or any other conventional or yet-to-be developed connections and/or connection techniques. As shown in the figures, all of the second housing **60** may be fabricated as a unitary, monolithic component, which may be moveably connected to the first portion **12** and second portion **14** of the first housing **10** as set forth above herein. In some embodiments a portion or all of the second housing **60** is transparent and/or translucent to permit a user to view through the transparent or translucent portion of the second housing **60** into the trap chamber **70**. In some embodiments, the first housing **10** is molded from plastics such as, for example, polypropylene, polyethylene, elastomer, copolymer, impact modified copolymer, polystyrene, combinations thereof, or the like, using conventional molding techniques such as, for example, injection molding, blow molding, compression molding, extrusion molding, laminating, reaction injection molding, matrix molding, rotational molding (or rotomolding), 3D printing, combinations thereof, or the like. The second housing **60** and/or its features and components may comprise any number of configurations, sizes, shapes, components, designs, and/or other materials (e.g., composites, metals, etc.).

[0041] If a user places a force on (e.g., pushes on) the second housing **60** toward the first housing **10** (e.g., downwardly or inward toward the reservoir **70** as indicated by arrow B in FIGS. **4** and **5**), the tabs **17**, **20** of the first housing **10** disengage and move out of the respective detents **66**, **68** of the second housing, permitting the second housing **60** to move from the first position (e.g., FIGS. **1**, **4**, and **5**) to the second position (e.g., FIG. **14**), closing the inlet **72** and trapping any insect(s) that is within the trap chamber **70** as set forth above herein. This engagement between the first and second housings permits a movable or slideable engagement between the housings, which allows one or more insects to enter the trap chamber and then a user to trap and prevent the one or more insects from exiting the trap **1**. It is understood that first and second housings may be engaged and/or moveably engaged with each other in any number of configurations and designs, including but not limited to, tongue and groove, tab and slide channel, hinged connected, pivoted connection, or rotational connection (e.g., dial connection).

[0042] In the embodiment shown, the floor **100** of the trap chamber **70** is formed by a substrate **110**. An adhesive **120** may coat a portion or all of a surface (e.g., upper surface **112**) of the substrate **110**. In some embodiments, more than one surface may be partially or completely coated with an adhesive. The substrate **110** is adhered via adhesive **120** to a bottom surface **19** of the bottom wall **16** of the first housing **10** opposite the trap chamber **70**. When connected as such to the first housing, the substrate **110** places and thus provides an adhesive coated substrate as the floor **100** in each of the channels **26** and thus the trap chamber **70**. The substrate **110** may include, temporarily, a protective substrate (e.g., wax paper) that is removably connected to the substrate via the adhesive **120** to cover and protect the adhesive. In some embodiments, the substrate **110** includes a commercially available glue card. The substrate **110** may be unconnected from the first housing **10** and then, after purchase, a user may peel off the protective sheet and adhere the substrate **110** to the first housing **10** as described above herein.

[0043] As shown in FIG. **10**, the bottom wall **16** may comprise a thickness (h). In some embodiments, when the substrate **110** is connected to the bottom surface **19** of the bottom wall **16**, the adhesive **120** may be configured to enter the plurality of sub-chamber apertures **23** such that the adhesive is at the same or substantially the same vertical elevation as the top surface of the bottom wall **16**. In such embodiments, the thickness of the adhesive **120** is equal to or greater than (h). Accordingly, one or more insects, being lured by the attractant (e.g., one or more attractants) disposed in the reservoir **40**, may enter the trap chamber **70** via the inlet **72** and eventually crawl onto the adhesive, trapping the one or more insects within the trap chamber **70**.

[0044] In some embodiments, the adhesive **120** may be configured to not enter or only partially enter the plurality of sub-chamber apertures **23** such that the adhesive is below the vertical elevation of the top surface of the bottom wall **16**. In such embodiments, the thickness of the adhesive **120** is less than (h), which creates a "pit" that the insects may fall into as they enter the trap **1** via the inlet **72**. Accordingly, one or more insects, being lured by the attractant (e.g., one or more attractants) disposed in the reservoir **40**, may enter the trap chamber **70** via the inlet **72** and eventually crawl and fall into one of the sub-chamber apertures. In such an embodiment, the one or more insects may not only be trapped by the pit created by this design, but the adhesive as well.

[0045] The same applies to the embodiments, wherein the bottom wall **16**, rather than a substrate **110**, comprises the floor **100** of the trap chamber **70**. In some embodiments, the floor **100** (i.e., the bottom wall **16** within the trap chamber **70**) may be at the same vertical elevation as the bottom wall at the inlet **72** into the trap chamber or at some vertical elevation below the bottom wall at the inlet **72** into the trap chamber. Similarly, these embodiments may or may not include adhesive. If they include adhesive, these embodiments may be configured such that the adhesive's upper surface is at a vertical elevation below, the same, or above the vertical elevation of an upper surface **9** of the bottom wall **16** at the inlet into the trap chamber. Accordingly, one or more insects, being lured by the attractant (e.g., one or more attractants) disposed in the reservoir **40**, may enter the trap chamber **70** via the inlet **72** and eventually crawl toward and get captured in the adhesive and/or fall into a pit created by the portion of bottom wall **16** that is at a lower vertical elevation, i.e., a pit. In such an embodiment, the one or more insects may be trapped by adhesive alone, the pit alone, or a combination of both.

[0046] Referring to FIG. **15**, as an example, another embodiment of trap **1** is partially shown. It is understood that this trap may include one or more of the components and/or features of the embodiments of trap **1** shown and described above herein. As shown, trap **1** may include the first housing **10**, the second housing **60** moveably engaged to the first housing, the trap chamber **70** defined by the first and second housings, and the inlet **72** providing a passage into the trap chamber **70** as shown and described herein. However, in this embodiment, the first housing includes one or more upward sloped portions **160**, creating a pit **162**, within the trap chamber **70**. As one or more insects enter the trap chamber **70**, the one or more insects may progress up the sloped portion **160** until the one or more insects reaches the peak **164** and then the one or more insects may fall into the pit **162**. In some embodiments, the pit **162** may also include an adhesive on one or more of its wall (e.g., bottom wall), providing an additional method of trapping the insect over-and-above the pit.

[0047] In some embodiments, the trap may include insect retention mechanisms in addition to or in place of the adhesive (e.g., adhesive **120**) and/or pit (e.g., pit **162**) to capture and/or arrest insects. These insect retention mechanisms may be used with the attractants herein. Illustrative insect retention mechanisms that may be included in one or more embodiments of the trap herein include, but are not limited to, histamine, diatomaceous earth, amorphous silica, biological control agents (e.g., pathogenic fungi), chemical toxin/insecticide, combinations thereof, and/or the like. In one example, the pit **162** or trap chamber floor **100** may include diatomaceous earth disposed therein. In another example, the pit **162** or trap chamber floor **100** may include amorphous silica. In another example, the pit **162** or trap chamber floor **100** may include a chemical toxin/insecticide. Other examples may include any combination of these retention mechanisms or similar ones. The diatomaceous earth and/or amorphous silica may injure and/or desiccate the insects such as, for example, bed bugs, and thus reduce the chance of the insects exiting the trap. In some embodiments, the trap may include histamine(s) disposed along (e.g., coated, sprayed, etc.), within, and/or impregnated within the first housing **10**, second housing **60**, the trap chamber **70**, and/or a cellulose surface. The histamine may also be disposed upon or incorporated into the adhesive. For example, a cellulose-based substrate (e.g., a “matt”) may be impregnated with histamine. The histamine-impregnated substrate may be placed on the floor **100** of the trap chamber or in the pit **162** of the trap chamber. Histamine may be used alone or in combination with the adhesive, pits, and/or the other insect retention mechanisms, and/or the attractants herein. It is understood that the other components of the trap (e.g., floor **100**) may be impregnated with histamine.

[0048] In some embodiments, the trap **1**, the first housing **10**, and/or the substrate **110** may include loop and hook and/or other mechanisms configured to connect (e.g., fixedly or removably) the trap to a bed, bed linens, carpet, and/or other materials. In some embodiments, the trap **1** (e.g., the first housing **10**) may be configured to comprise a dark color to attract insects such as, for example, red or black. In some embodiments, the color only has to appear dark to the insect you are trying to attract, capture, and/or arrest. For example, the color red appears dark to a bed bug. In some embodiments, one or more outer surfaces of the trap **1** (e.g., outer surfaces of the first housing **10** and/or the second housing **60**) are configured to be smooth (e.g., very smooth) to discourage insects from staying on the outer surfaces of the trap. In some of these embodiments, this smooth or very smooth outer surface(s) may increase the likelihood of the insects entering the trap. The material for the housings may be selected and/or fabricated to provide one or more of the trap's outer surfaces with a smooth or very smooth surface. In some embodiments, one or more of the trap's outer surfaces, trap chamber's surfaces, and/or pit's surfaces may be coated with a material that provides a smooth or very smooth surface.

[0049] In some embodiments, the floor **100** and/or the upper surface **9** of the bottom wall **16** (e.g., at the inlet **72**) may be configured to provide the insects grip (e.g., optimal or greater grip) which may encourage the insects to enter the trap **1**. In some embodiments, the floor **100** and/or the upper surface **9** may include a plurality of cilia extending from such surface(s) to provide a grip (e.g., optimal grip) for insects entering the trap. It is understood that other mechanisms may be used to improve the grip of the insects on such surfaces such as, for example, low durometer materials, etc.

[0050] In some embodiments, a kit for an insect trap may include one or more of the features of the traps shown and described herein in assembled, partially assembled, or unassembled configurations. For example, a kit for an insect trap **1** may include a first housing including a first surface, a reservoir disposed within the first housing, a protrusion disposed within the reservoir and extending inward from an interior surface of the first housing, and an actuator disposed upon an exterior surface of the first housing and connected to the protrusion such that when the actuator is moved, the protrusion moves within the reservoir; a second housing moveably engaging the first housing; an attractant **150** configured to attract insects and to be disposed within the reservoir; a substrate comprising an adhesive disposed upon at least a portion of a surface of the substrate, the substrate configured to adhere to the first surface of the first housing; a trap chamber formed by the first housing, second housing, and substrate when the substrate is adhered to the first surface of the first housing; and an inlet providing a passage into the trap chamber; wherein the second housing is movable between a first position where the inlet is open and a second position where the inlet is closed. In some embodiments, the kit may include one or more of the mechanisms shown and described herein to capture and/or arrest the insects that enter the trap. The attractant will be described below herein.

[0051] In some embodiments, the floor **100** of the trap chamber **70** is formed by the bottom wall **16** of the first housing. In such embodiments, the bottom wall **16** does not include the trap chamber aperture **21** and thus does not include the plurality of sub-chamber apertures **23**. An adhesive (not shown) may coat a portion or all of a surface (e.g., the upper surface **9**, opposite bottom surface **19**, of the first housing **10**) of the substrate **110**. In some embodiments, more than one surface of the first housing **10** may be partially or completely coated with an adhesive.

[0052] The trap **1** may include an attractant **150** partially or completely disposed within the reservoir **70**. In some embodiments, the attractant **150** comprises an attractant composition configured to attract or lure insects, including but not limited to bed bugs, bat bugs, swallow bugs, poultry bugs, and *Hesperocimex coloradensis*, the first package (or inner package) **152** enclosing the attractant composition therein, and the second package (or outer package) **154** enclosing the inner package **152**, and thus the attractant composition, therein.

[0053] In some embodiments, the attractant composition comprises a pheromone composition of one or more pheromone components and/or one or more other ingredients (e.g., a pheromone blend) as shown and described in PCT/CA2014/051218, filed Dec. 16, 2104, which is incorporated herein by reference.

[0054] Some aspects of the invention pertain to compositions for attracting and/or arresting bed bugs, such as the common bed bug, *Cimex lectularius*, and the tropical bed bug, *C. hemipterus*. In some embodiments, the composition may comprise pheromone components isolated from the exuviae or faeces of the bed bug *Cimex lectularius*, or synthetic equivalents of such compounds. In some embodiments the composition comprises histamine. In some embodiments the active ingredient of the composition may essentially consist of histamine. In some embodiments, histamine is provided as a base. In some embodiments the composition comprises histamine and a blend of volatile compounds comprising sulfides, aldehydes and ketones. In some embodiments the blend of volatile compounds comprises one or more of dimethyl disulfide, dimethyl trisulfide, (E)-2-hexenal, (E)-2-octenal, and 2-hexanone. In some embodiments the active ingredients of the composition may essentially consist of histamine and one or more of dimethyl disulfide, dimethyl trisulfide, (E)-2-hexenal, (E)-2-octenal, and 2-hexanone. In some embodiments the composition may comprise or essentially consist of histamine, dimethyl disulfide, dimethyl trisulfide, (E)-2-hexenal, (E)-2-octenal, and 2-hexanone.

[0055] In some embodiments the compositions described herein may include one or more additional active ingredients such as butanal, pentanal, hexanal, benzaldehyde, benzyl alcohol, acetophenone, verbenone, ethyl octanoate, methyl octanoate, pentyl hexanoate, dimethylaminoethanol, N-acetylglucosamine, 3-hydroxykynurenine O-sulfate, L-valine, L-alanine, octanal, nonanal, decanal, (E,E)-2,4-octadienal, (E,Z)-2,4-octadienal, benzyl acetate, (+)-limonene, (−)-limonene, 6-methyl-5-hepten-2-one (sulcatone), geranylacetone, carbon dioxide, 1-octen-3-ol, L-carvone, L-lactic acid, proprionic acid, butyric acid, valeric acid, oleic acid, palmitic acid, stearic acid, linoleic acid, lauric acid, capric acid, myristic acid, androstenone, 3-methyl indole, 1-docosanol, pentadecanoic acid, squalene, cholesterol, and 2,2-dimethyl-1,3-dioxolane-4-methanol.

[0056] In some embodiments, the compositions described herein may be formulated as a granule, powder, dust, paste, gel, suspension, emulsion or liquid solution.

[0057] Certain aspects of the invention pertain to methods for attracting and/or arresting bed bugs, such as the common bed bug, *Cimex lectularius*, and the tropical bed bug, *C. hemipterus*, in a desired location. In some embodiments the methods comprise providing a composition as described herein at the desired location. In some embodiments the desired location can be in, on or near a bed bug control device which is located in any structure occupied by a human or animal, such as one or more buildings, vehicles, dwellings and/or living spaces in a residential, institutional, commercial, industrial, governmental, or agricultural setting, in which bed bugs were present, are present or are suspected to be present. In some embodiments the suitable bed bug control devices include detectors, monitors and traps.

[0058] In some embodiments the compositions may be formulated as a lure, such as a slow release lure, and provided in, on or near a bed bug control device. In some embodiments the compositions may be provided in a slow release device provided in, on or near a bed bug control device. In some embodiments, certain components of the composition may be provided in separate slow release devices and/or other release devices (e.g., normal or standard release). For example, in some embodiments the deployment mechanism for histamine may be an absorbent material (e.g., an absorbent mat such as a cellulose or cellulose-based substrate (e.g., mat)) impregnated with histamine, and the slow release device for the volatile compounds as described herein or the volatile compounds and the one or more additional compounds as described herein, may be a gas-permeable sealed reservoir (e.g., a pheromone-permeable, sealed plastic reservoir). In some embodiments the components of the composition may be provided in the same slow release device.

[0059] In some embodiments the compositions may be combined with a source of heat, carbon dioxide and/or a pesticide that is lethal to bed bugs. As set forth above, in some embodiments, the pesticides may be diatomaceous earth, amorphous silica, biological control agents (e.g., pathogenic fungi), or the like.

[0060] Certain aspects of the invention pertain to the uses of compounds and compositions for attracting and/or arresting bed bugs. In some embodiments, histamine may be used to arrest bed bugs. In some embodiments, the compositions as described herein are used to attract and/or arrest bed bugs.

[0061] In some embodiments, the second package **154** encloses the first package **152** and is configured to prevent one or more ingredients of the attractant composition and gas such as, for example, an active ingredient, from evaporating by volatizing through a wall of the first package **152** and a wall of the second package **154**. In some embodiments, the second package **154** includes an exterior wall that is impermeable to gas, thus preventing the attractant composition from releasing from the first and/or second packages **152**, **154**. As an example, the second package **154** may comprise a sealed pouch fabricated from a laminate including a high barrier polyethylene terephthalate layer, an aluminum foil layer, and a polyethylene coextruded film.

[0062] In some embodiments, the first package **152** may be configured to permit one or more ingredients of the attractant composition such as, for example, an active ingredient, to permeate a wall of the first package **152** (e.g., evaporating by volatizing through the wall). For example, the first package **152** may include one or more walls fabricated from one or more materials that is permeable to gas. As an example, the one or more walls of the first package **152** may comprise polyvinylchloride (PVC). In one embodiment, the first package **152** is a hollow cylindrical tube made of PVC or similar gas permeable materials such as, for example, polyethylene. In some embodiments, the tube includes opposite ends that are both sealed closed (e.g., heat sealed), thus enclosing the attractant composition therein.

[0063] In some embodiments, the first package **152** includes an exterior wall that is configured to release the attractant composition through the exterior wall at a release rate from about 5 μg per day to about 500 μg per day. In some embodiments, the first package **152** includes an exterior wall that is configured to release the attractant composition through the exterior wall at a release rate from about 15 μg per day to about 400 μg per day. In some embodiments, the first package **152** is gas permeable such that the first package releases the attractant composition at a release rate of greater than about 15 μg per day on the 20th day after the opening of the first package or, in some embodiments, greater than about 20 μg per day on the 20th day after the opening of the first package.

[0064] In one example, the first package **152** is fabricated from polyvinylchloride (PVC) tube having heat-sealed ends. An attractant composition comprising: 19.98, % w/w, 2-hexanone (591-78-6); 19.98, % w/w, (E)-2-hexenal (6728-26-3); 19.98, % w/w, (E)-2-octenal (2548-87-0); 19.98, % w/w, dimethyl disulfide (624-92-0); 19.98, % w/w, dimethyl trisulfide (3658-80-8); 0.1, % w/w, Sudan B black (4197-25-5) is blended first to form an insect pheromone (bed bug pheromone) composition. This insect pheromone composition is then mixed and/or dissolved in mineral oil (8042-47-5) to form that attractant composition (e.g., the attractant composition having a concentration of 5.0, % w/w, insect pheromone composition and 95.0, % w/w, mineral oil). This combination of attractant composition within the PVC tube had a release rate from about 20 μg per day to about 350 μg per day. In some embodiments, the Sudan B black is removed and the concentration of the remaining five components/ingredients may be increased from 19.98, % w/w, each to 20, % w/w, each.

Test Method

[0065] The test method used to determine the release rate of the attractant composition from the first package (e.g., gas permeable package) includes the following steps. [0066] 1) A total number of attractant first packages (i.e., a gas permeable package having one or more attractant compositions enclosed therein (e.g., an insect lure package)) is determined to be included in the test group. Any number can be included in the test group, but the test group should include at least 20 attractant first packages, preferably 50. [0067] 2) Measuring the total mass of the first attractant packages in the test group at the same time, immediately after each attractant first package has been opened (e.g., removed from a gas impermeable package or gas impermeable storage container and thus exposed to the atmosphere). This provides the total mass (e.g., in micrograms (μg)) of the attractant first packages (e.g., total mass of a minimum of 20, preferably 50, attractant first packages) at the starting point (e.g., T.sub.0=0 days). [0068] 3) After a first period (T.sub.1) in days (each day equals a 24 hour period) has transpired from the point the attractant first packages were exposed to the atmosphere, the mass of all the attractant first packages (e.g., minimum 20, preferred 50) is measured in total again. It is preferred that the first period equals three (3) days (e.g., T.sub.1=3 days). However, the first period can be any period of days, but preferably more than one (1) day. [0069] 4) The total mass of the attractant first packages in the test group at step 2 (e.g., T.sub.1=3 days) is then subtracted from the total mass measured at step 1 (T.sub.0=0). This provides the total mass loss (e.g., μg) for the attractant first packages in the test group after the first period (e.g., T.sub.1=3 days). [0070] 5) This total mass loss at T.sub.1 is then divided by the number of units in the first time period (T.sub.1=3 days). Thus, the total mass loss is divided by three (3) days to provide the average total mass loss per unit of time (e.g., μg per day). [0071] 6) This average total mass loss per day is then divided by the total number of attractant first packages in the test group (e.g., a minimum of 20, preferably 50) to provide an average total mass loss per day per attractant first package (e.g., μg per day per attractant first package). [0072] 7) After a second period (T.sub.2) in days has transpired from T.sub.1(e.g., T.sub.2=3 days), the mass (e.g., μg) of all the attractant first packages in the test group is measured in total again. The second period can be any period of days, but preferably more than one (1) day. [0073] 8) The total mass of the attractant first packages in the test group measured at step 7 (T.sub.2) is then subtracted from the total mass measured of the attractant first packages at step 3 (T.sub.1). This provides the total mass loss (e.g., μg) for the attractant first packages in the test group from the first time period (T.sub.1) through the second period (e.g., T.sub.2=3 days). [0074] 9) This total mass loss at T.sub.2 is then divided by the number of units in the second time period (e.g., T.sub.2=3 days). Thus, the total mass loss is divided by three (3) days to provide the average total mass loss per unit of time (e.g., μg per day) during this period. [0075] 10) This average total mass loss per unit of time (per day) is then divided by the total number of attractant first packages in the test group to provide an average total mass loss per day per attractant first package (e.g., μg per day per attractant first package) during the second time period (T.sub.2). [0076] 11) After a n.sup.th period (T.sub.n) in days has transpired from T.sub.n-1, the mass (e.g., μg) of all the attractant first packages in the test group is measured in total again. [0077] 12) The total mass of the attractant first packages in the test group measured at step **11** (T.sub.n) is then subtracted from the total mass measured of the attractant first packages at (T.sub.n-1). This provides the total mass loss (e.g., μg) for the attractant first packages in the test group from the n−1 period (T.sub.n-1) through the n.sup.th period (T.sub.n). [0078] 13) This total mass loss at T.sub.n is then divided by the number of units of time (e.g., days) in the n.sup.th period (T.sub.n) to provide the average total mass loss per unit of time (e.g., μg per day) during this period. [0079] 14) This average total mass loss per unit of time (e.g., per day) is then divided by the total number of attractant first packages in the test group to provide an average total mass loss per day per attractant first package (e.g., μg per day per attractant first package) during the n.sup.th period (T.sub.n). [0080] 15) Steps 11-14 are repeated to calculate the average total mass loss per day per attractant first package through day twenty (20). [0081] 16) Steps 11-14 may be repeated to calculate the average total mass loss per day per attractant first package through any other number of units of time (e.g., 25 days, 30 days, etc.).

[0082] In some embodiments, a method of capturing one or more insects may include peeling a protective sheet from an adhesive layer on a substrate, connecting the substrate to a bottom surface of a first housing via the adhesive such that the adhesive of the substrate forms a floor of a trap chamber of a trap, applying a force to an actuator to open a package containing an attractant composition within the trap and causing the attractant composition to release and attract insects, placing the trap in a location (e.g., a bed, floor, etc.) with an inlet of the trap in its open position, and allowing the trap to sit in the location for a period of time. After a period of time has transpired, the method may further include viewing into a window (e.g., lifting or peeling cover **80** back from transparent portion of second housing **60**) to determine whether any insects, having been lured into the trap chamber via the attractant composition, are captured within the trap chamber via any of the trap embodiments shown and described herein (e.g., pit, adhesive, histamine, and/or combinations thereof). If insects are captured within the trap chamber, the method may include a user closing the inlet to the trap chamber (e.g., applying a force to second housing **60** to move it from the first position to the second position). Once closed, the method may include removing the trap from the location. It is understood that any of these steps may be removed, the order rearranged, and/or other steps added to such method. Additionally such method and one or more of its steps may be applied to any of the other trap embodiments shown and described herein.

[0083] In operation, some embodiments may be activated by applying a force to the button **46** which causes the protrusion to move inward within the reservoir **40** such that it pierces the second package **154** of the attractant **150** disposed with the reservoir **40** of the trap **1**. Once the force is released, the button **46** and protrusion retract back to their original positions, thus moving the protrusion away from the puncture hole it created within the second package **154**. Upon this puncturing of the second package **154**, the one or more ingredients of the attractant composition begin to volatilize and release through the first package **152**, attracting and luring one or more insects (e.g., bed bug(s)) into the inlet **72** of the trap **1**. Once inside the trap chamber **70**, the one or more insects may be trapped and/or captured by one or more of the trap embodiments (techniques/methods) shown and disclosed herein (e.g., adhesive, pit, histamine, pesticide, or combinations thereof). Once captured, a user may view into the trap chamber **70** through a transparent portion of second housing **60** (which may include lifting or peeling cover **80** from the second housing **60**). A user may apply a force to the second housing **60** to move it from the first position to the second position, thus closing the inlet **72**.

[0084] Referring to FIGS. **17** and **18**, an embodiment of an insect trap **200** is shown. The insect trap **200** generally may include a housing **10**, a reservoir **40** at least partially disposed within or defined by the first housing, a protrusion **42** disposed within the reservoir, an actuator **44** connected to and configured to actuate the protrusion, a substrate **255**, a first cover **258**, a trap chamber **270** partially formed by the housing and the substrate, and an inlet **272** providing an entrance into the trap chamber. The trap chamber may include a floor **300**. The inlet **272** is open, permitting insects to enter and/or exit the trap chamber **270**. The trap **200** may include a longitudinal axis (L-L′) as shown.

[0085] As shown in FIGS. **17** and **18**, the substrate **255** comprises a continuous wall **257** defining a first plurality of interior channels **226**. The substrate **255** may define a second plurality of interior channels **226′** on the opposite side of the continuous wall **257** from the first plurality of interior channels **226**. In the illustrated embodiment, the channels **226** are open in a direction away from the floor **300**, and the channels **26′** are open in a direction towards the floor **300**. Each channel **226**, **226′** is disposed between two adjacent angled portions **263** of the continuous wall **257** or an angled portion of the continuous wall and the third upper housing wall **38***c*. Between each of the adjacent angled portions **263** of the continuous wall **257** are arcuate or rounded portions **265** that define a floor of the channels **226**, **226′**. While the channels **226**, **226′** in the illustrated embodiment have a U-shaped cross-section, the cross-section may have another shape. The plurality of interior channels **226**, **226′** may have any length and height. In some embodiments, the height of each channel **226**, **226′** may be about 0.08 in. The plurality of channels **226** forms a plurality of sub-chambers within the trap chamber **270**. In some embodiments, the substrate **255** is configured to prevent insects from entering the plurality of channels **226′**. For example, in the illustrated embodiment, the ends of the channels **226′** are pinched together or flattened, i.e., closing off the entrance(s) to the channels **226′**, so that insects may not enter. In some embodiments, the substrate **255** also includes a bottom wall **259** coupled to the continuous wall **257**. The substrate **255** is coupled to the housing **10** such as, for example, using adhesive to adhere the substrate **255** to the housing **10**. In some embodiments, a portion of the substrate **255** extends under the housing **10**. In other words, the housing **10** may be positioned on the substrate **255**. Where the continuous wall **257** extends under the housing **10**, the channels **226**, **226′** may be closed (e.g., flattened) to prevent insects from entering the space below the housing **10**.

[0086] In some embodiments, the substrate **255** is made from paper or fiber. As an example, the substrate **255** may be made of single face corrugated paper or paperboard. The substrate **255** and/or its features and components may comprise any number of configurations, sizes, shapes, components, designs, and/or other materials (e.g., composites, metals, etc.). The substrate **255**, and one or more of its components, may be fabricated as a unitary, monolithic component. In some embodiments, the substrate, and one or more of its components, may be fabricated as multiple components that may be connected (e.g., fixedly or removably) to each other, using any connection method described herein and/or any other conventional or yet-to-be developed connections and/or connection techniques.

[0087] In the embodiment shown, the ceiling **302** of the trap chamber **270** is formed by the first cover **258**. An adhesive **320** may coat a portion or all of a surface (e.g., a lower surface **312**) of the first cover **258**. In some embodiments, more than one surface may be partially or completely coated with an adhesive. In some embodiments, a portion or all of the first cover **258** may be transparent and/or translucent to permit a user to view through the transparent or translucent portion of the first cover **258** and adhesive **320** into the trap chamber **270**. The first cover **258** is adhered via adhesive **320** to the continuous wall **257** of the substrate **255**. When connected as such to the substrate **255**, the first cover **258** places and thus provides an adhesive coated substrate as the ceiling **302** in each of the channels **226** and thus the trap chamber **270**. Additionally, the first cover **258**, depending on the material used, may increase the stability or structural strength of the substrate **255**. In some embodiments, the first cover **258** is made from plastic such as, for example, polyethylene terephthalate (e.g., Mylar®). An example first cover is a clear sticky panel trap including No-Mess Adhesive™ provided by Alpha Scents, Inc. In some embodiments, the first cover **258** may include, temporarily, a protective substrate (e.g., wax paper) that is removably connected to the substrate via the adhesive **320** to cover and protect the adhesive. The first cover **258** may be unconnected from the substrate **255** and then, after purchase, a user may peel off the protective sheet and adhere the first cover **258** to the substrate **255** as described above herein.

[0088] An inlet **272** is defined between the upper surface of the continuous wall **257** of the substrate **255** and the bottom surface of the first cover **258** as shown in FIGS. **17** and **18**. For example, an insect may enter the inlet **272** and walk onto an arcuate portion of the wall **257**. The insect may continue walking through the channel **226** and become adhered to the adhesive **320** on the bottom surface of the first cover **258** (i.e., near the ceiling **302** of the trap chamber **270**).

[0089] In some embodiments, a method of capturing one or more insects may include peeling a protective sheet from an adhesive layer on a cover, connecting the substrate to a top surface of a substrate via the adhesive such that the adhesive of the cover forms a ceiling of a trap chamber of a trap, applying a force to an actuator to open a package containing an attractant composition within the trap and causing the attractant composition to release and attract insects, placing the trap in a location (e.g., a bed, floor, etc.), and allowing the trap to sit in the location for a period of time. After a period of time has transpired, the method may further include viewing into a window (e.g., through the cover **258** or lifting or peeling a separate cover (e.g., cover **80**) back from transparent portion of the cover **258**) to determine whether any insects, having been lured into the trap chamber via the attractant composition, are captured within the trap chamber via any of the trap embodiments shown and described herein (e.g., adhesive, histamine, and/or combinations thereof). If insects are captured within the trap chamber, the method may include removing the trap from the location. It is understood that any of these steps may be removed, the order rearranged, and/or other steps added to such method. Additionally such method and one or more of its steps may be applied to any of the other trap embodiments shown and described herein.

[0090] The invention can be further understood by reference to examples, of which summaries and detailed descriptions follow. These examples are provided by way of illustration and are not meant to be limiting.

Example 1

Maintaining a Colony of Common Bed Bugs for Production of Exuviae and Use in Bioassays

[0091] A colony of common bed bugs was kept in an insectary at 22-24° C., ambient relative humidity, and a photoperiod of 10 hours dark to 14 hours light. To collect pheromone for extraction, isolation, and identification, the colony was increased from 2,400 to 6,000 bed bugs and held at the higher level for 18 months.

[0092] Approximately 150 bed bugs were kept in each of 40 50-ml jars. Each jar was fitted with a piece of cardboard (2×2 cm) at the bottom and a strip (2×5 cm) of corrugated cardboard diagonally across the jar. The jar was covered with a plastic lid perforated with small holes for ventilation.

[0093] Each bed bug was allowed to feed once per month on a human volunteer. At 1,500 bed bugs per week for 30 months, this amounts to 180,000 individual feedings. Jars with bed bugs to be fed were covered with fine mesh and pressed against the volunteer's forearm so that the bed bugs could feed through the mesh. After feeding, nymphal bed bugs moult, shedding their exuvia in the process. Each exuvia of a 5th instar nymph weighs about 0.07 mg. Collecting exuviae of 1,200 5th instar nymphs (20% of the entire colony) per month, resulted in a harvest of 84 mg (1,200×0.07 mg) of exuviae per month for a total of 1,512 mg of exuviae. This was the starting material for extraction, isolation, and identification of the aggregation pheromone.

Example 2

General Experimental Design to Investigate the Response of Bed Bugs to Test Stimuli

[0094] Bioassays were run in dual-choice olfactometers and in large Plexiglass arenas (FIG. **19**). Dual-choice olfactometers consisted of two lateral Pyrex® glass Petri dishes, connected to a central dish (all dishes 3×9 cm inner diameter) via a Pyrex® glass tube (2.5 cm long×2 cm inner diameter). The dishes in this olfactometer mimic the natural still-air shelters in which bed bugs spend the day. Prior to the start of bioassays, a disc of paper toweling (9 cm diameter) was placed into each dish and a strip of paper toweling (2.4×0.6 cm) was inserted into the connecting glass tubing to provide traction for walking bed bugs. In addition, a piece of filter paper (2×3 cm; Whatman) was placed into each lateral dish and covered with a piece of cardboard (2×2 cm) as a refuge for bioassay insects.

[0095] Treatment and control stimuli were randomly assigned to each lateral dish. Olfactometers were enclosed in opaque plastic bins to prevent light from entering and affecting the insects' responses. For each replicate, a single 3.sup.rd, 4.sup.th, or 5.sup.th instar *C. lectularius* nymph was released into the central chamber of the olfactometer, which was then covered with a glass plate to prevent escape. The single bed bug in each olfactometer was then allowed to explore all chambers. Each insect was released into an olfactometer at the end of the 14-h photophase allowing it to explore the chambers during 10 h of darkness, and to come to rest in one of the shelters during 10 h of light. After this 20-h period, the insect's position within the olfactometer was recorded. Any insects not found in a lateral chamber were recorded as non-responders. All experimental replicates were run at 24±2° C. and 30-60% RH. Olfactometers were washed with Sparkleen detergent (Fisher Scientific Co, Pittsburgh, PA, USA), and were oven-dried (Precision, Winchester, VA, USA) between each bioassay.

[0096] The large two-chamber Plexiglass arena (180 cm long×12 cm high×13 cm wide) was designed with a central divider (180 cm long×5.5 cm high) to accommodate two pieces of wood (162 cm long×3.8 cm wide×0.8 cm high), each piece for testing the response of a single bed bug to a pheromone-baited shelter or a control shelter (see insert in FIG. **19**) which were randomly assigned to either end of the wood. As some bed bugs succeeded in crossing over the divider, from one chamber to the other, all experimental replicates used only one chamber of each arena. For each replicate, a single bed bug was placed in the center of the wood piece just prior to the end of the 14-h photophase, and its position was scored the next morning after the onset of the photophase.

Example 3

Evidence that Bed Bug Exuviae (Cuticle Shed During Moulting) Induce Arrestment of Foraging Bed Bugs: Effect of Number and Age of Exuviae Tested

[0097] Experiment 1 tested whether 50 exuviae of 5.sup.th instar nymphal bed bugs (1 exuvia=0.08 mg) induce arrestment of foraging 5.sup.th instar nymphs. Given a strong arrestment response of nymphs to 50 exuviae (see Table 1), follow-up Experiments 2-4 tested whether fewer numbers of exuviae would suffice to induce arrestment responses. Experiment 5 explored whether exuviae which were aged at room temperature for 2 months are still effective in inducing arrestment of bed bugs.

[0098] In Experiments 1-4, 50, 10, 2 or 1 exuviae were unexpectedly all equally effective in inducing arrestment of bed bugs (Table 1). In Experiment 5, exuviae after 2 months of storage at room temperature still induced arrestment of bed bugs.

[0099] Table 1 illustrates the effect of number and age of exuviae on the response of bed bugs in the three-dish, dual-choice olfactometer in Experiments (Exp.) 1-5.

TABLE-US-00001 No. arrested No. arrested Exp. No. in baited in control No. non- no. exuviae chamber chamber responders 1 50 12 0 0 2 10 12 0 0 3 2 21 1 2 4 1 10 2 0 5 2 (stored 1 month) 12 0 0

Example 4

Efficacy of Organic Solvent to Extract Pheromone from Exuviae

[0100] Experiments 6-10 tested the efficacy of different organic solvents (hexane, ether, dichloromethane, acetonitrile, methanol) to extract pheromone from exuviae. Each olfactometer experiment tested 6-exuviae equivalents of extracts, i.e., the amount of material, possibly including pheromone, which could be extracted from a total of six exuviae.

[0101] In Experiments 6-9, hexane, ether, dichloromethane and acetonitrile were not effective in extracting pheromone from exuviae and inducing an arrestment response of 3.sup.rd to 5.sup.th instar nymphs (Table 2). In Experiment 10, the methanol extract of exuviae induced a strong arrestment response (Table 2), indicating that the bed bug arrestment pheromone component was present in the methanol extract.

[0102] Table 2 illustrates the effect of organic solvent used for extraction of exuviae on the response of bed bugs in the three-dish, dual-choice olfactometer in Experiments (Exp.) 6-10.

TABLE-US-00002 No. arrested No. arrested Exp. Solvent in baited in control No. non- no. tested chamber chamber responders 6 Hexane 6 6 0 7 Ether 3 5 4 8 Dichloromethane 1 7 3 9 Acetonitrile 4 7 2 10 Methanol 10 1 0

Example 5

Isolation of the Arrestment Pheromone Component

[0103] Presence of the arrestment pheromone component in the methanol extract of exuviae (Table 2; Exp. 10) indicated that it had a molecular structure of significant polarity. To isolate the pheromone component for structural elucidation, exuviae were extracted in sequence in organic solvents of increasing polarity (hexane, ether, dichloromethane, acetonitrile, and methanol). Consequently, the final methanol extract contained primarily polar compounds. This methanol extract was then fractionated through silica gel (0.6 g) in a glass column (14 cm long×0.5 cm inner diameter). After the silica was pre-rinsed with pentane, the methanol extract was applied, allowed to impregnate the silica gel, and then eluted with 5 consecutive rinses (2 ml each) of pentane/ether, with increasing proportions of ether [1) 100:0; 2) 90:10; 3) 80:20; 4) 50:50; 5) 0:100], followed by five consecutive rinses (1 ml each) of dichloromethane/methanol, with increasing proportions of methanol [1) 100:0; 2) 90:10; 3) 80:20; 4) 50:50; 5) 0:100]. The five dichloromethane/methanol fractions were then bioassayed in Experiments 11-15.

[0104] In Experiments 11-15 (Table 3), only the silica fraction with 50% methanol as eluent (Exp. 14), induced arrestment of bed bug nymphs.

[0105] Table 3 illustrates the effect of the solvent system on eluting the bed bug arrestment pheromone component(s) from silica gel. Note that only the test stimulus in Experiment 14, consisting of dichloromethane (CH.sub.2Cl.sub.2; 50%) and methanol (MeOH; 50%) as eluents, induced arrestment responses in bed bugs in the three-dish, dual-choice olfactometer.

TABLE-US-00003 No. arrested No. arrested Exp. Solvent in baited in control No. non- no. system chamber chamber responders 11 CH.sub.2C1.sub.2 (100%) 5 6 1 12 CH.sub.2C1.sub.2/MeOH (10%) 5 5 2 13 CH.sub.2C1.sub.2/MeOH (25%) 6 4 2 14 CH.sub.2C1.sub.2/MeOH (50%) 9 2 1 15 MeOH (100%) 4 2 6

[0106] This protocol for pheromone isolation was repeated in the same or slightly modified form several times. Each time, the responses of bioassay insects to silica fractions indicated that the arrestment pheromone component was present in a fraction eluted with 50% or 100% methanol. These results combined clearly revealed that the arrestment pheromone component is highly polar.

Example 6

Pheromone Identification: Micro-Analytical Treatments of Bio-Active Extract

[0107] To determine whether the polar arrestment pheromone component has an acid, amine, or alcohol functionality, methanol extracts of exuviae (see Table 2) were subjected to micro-analytical treatments with diazomethane (converts acids to esters) or acetic anhydride in pyridine (converts alcohols to esters) and then bioassayed in Experiments 16 and 17.

[0108] In Experiment 16 (Table 4), diazomethane-treated methanol extract of exuviae induced arrestment responses of bed bug nymphs, indicating that the diazomethane treatment did not alter the pheromone molecule and that this pheromone component does not likely have an acid functionality. Conversely, in Experiment 17, acetic anhydride-treated methanol extract of exuviae failed to induce arrestment of bed bug nymphs, indicating that the acetic anhydride treatment had altered the molecular structure of the pheromone component and that it had one or more hydroxyl and/or amine groups.

[0109] Table 4 illustrates the effect of micro-analytical treatments of pheromone extract on the arrestment responses of bed bug nymphs in Experiments 16 and 17. Note that the acetic anhydride treatment of methanol extract of exuviae altered the molecular structure of the pheromone component and thus failed to induce a significant arrestment response of bed bug nymphs in the three-dish, dual-choice olfactometer.

TABLE-US-00004 No. arrested No. arrested Exp. in baited in control No. non- no. Treatment chamber chamber responders 16 Diazomethane-treated 9 0 3 pheromone extract 17 Acetic anhydrite-treated 3 6 3 pheromone extract

Example 7

Pheromone Identification: Nuclear Magnetic Resonance Spectroscopy (NMR) of Bioactive Extract

[0110] To elucidate the molecular structure of the arrestment pheromone component, the .sup.1H NMR spectra of several methanol extracts of exuviae and feces were examined and compared. Analyses of the .sup.1H NMR spectra revealed several common components that were identified as L-valine, L-alanine, N-acetylglucosamine, histamine, dimethylaminoethanol, and 3-hydroxykynurenine O-sulfate.

[0111] Valine, alanine, N-acetylglucosamine, histamine, and dimethylaminoethanol were identified by comparison of the observed .sup.1H and .sup.13C NMR and mass spectrometric data with those reported previously for these compounds. In addition, authentic samples of L-valine, N-acetylglucosamine, histamine, and dimethylaminoethanol were purchased from commercial vendors and added to a crude methanol extract in separate experiments. In each of these additional experiments, the resonances observed in the .sup.1H NMR spectra recorded on the crude methanol extract and assigned to a specific component (i.e., L-valine, L-alanine, N-acetylglucosamine, histamine, and dimethylaminoethanol) were enhanced by the addition of an authentic sample of that component, confirming the identity and occurrence of these chemicals in the crude methanol extracts.

[0112] The structure of 3-hydroxykynurenine O-sulfate was proposed following comparison of specific resonances observed in the .sup.1H NMR spectra recorded on the crude methanol extracts to those reported previously for 3-hydroxykynurenine O-sulfate. Additionally, an authentic sample of 3-hydroxykynurenine O-sulfate was prepared from 3-hydroxykynurenine following protection of the carboxylic acid as a methyl ester and the amine function as a carboxybenzyl amide. Sulfation of the free alcohol using Me3N—SO.sub.3, followed by removal of the carboxybenzyl protecting group by hydrogenolysis and hydrolysis of the methyl ester provided an authentic sample of 3-hydroxykynurenine O-sulfate. The .sup.1D and .sup.2D NMR spectra (.sup.1H, COSY, HMQC, HMBC) recorded on the synthetic sample of 3-hydroxykynurenine O-sulfate were in complete agreement with the natural material present in the crude methanol extract.

[0113] Table 5 summarizes the relative amounts of each of these components in each of five individual methanol extracts, and Table 6 shows the biological activity of each extract in Experiments 18-22. Note that the highest response levels (Exp. 18-20) were achieved with the three extracts in which histamine and dimethaminoethanol were both present in appreciable amounts.

[0114] Table 5 illustrates the relative amount in mg of each of the common components found in the methanol extracts of bed bug exuviae and feces.

TABLE-US-00005

Compound	Extract No. 1	2	3	4	5
L-Valine	0.14	0.07	0.0	0.0	0.06
L-Alanine	0.0	0.0	0.0	0.0	0.0
N-Acetylglucosamine	0.0	0.0	0.0	0.25	0.0
3-Hydroxykynurenine O-sulfate	3.7	2.5	3.2	0.48	2.1
Histamine	4.7	1.9	8.2	0.0	0.0
Dimethylaminoethanol	3.0	0.7	0.9	0.0	0.0

[0115] Table 6 illustrates the responses of bed bugs in bioassays in the three-dish, dual-choice olfactometer of the five extracts analyzed for constituent components in Table 5.

TABLE-US-00006

Extract No.	Experiment No.	No. arrested in baited chamber	No. arrested in control chamber	No. non-responders
1	18	10	1	1
2	19	10	1	1
3	20	10	1	1
4	21	9	2	2
5	22	7	4	2

Example 8

Histamine: A Pheromone Component of Bed Bugs

[0116] With the highest response levels of bed bugs achieved with extracts in which histamine and dimethylaminoethanol were both present (Tables 5, 6), follow-up experiments were designed to test the response of bed bugs to authentic standards. In Experiments 23-25, increasing doses of dimethylaminoethanol and histamine at a ratio of 1:1 improved the bed bugs' arrestment response (Table 7). Experiments 26-28 then tested dimethylaminoethanol (20 μg) and histamine (20 μg), with histamine presented as a base, salt or base and salt. The results reveal that histamine only as a base elicits the bed bugs' arrestment response (Table 7). Consequently, all further experiments deployed histamine as a base.

[0117] Experiments 29-31 tested whether a 10-fold (from 20 μg to 200 μg) increase of dimethylaminoethanol or of histamine improves lure effectiveness. The results indicate that the 10-fold increase of histamine enhances the bed bugs' arrestment response (Table 7). Drawing on results of Experiments 29-31, Experiments 32-34 tested dimethylaminoethanol and histamine at a 1:10 ratio at doses of 2 μg:20 μg, 20 μg:200 μg, and 200 μg:2000 μg. The results show that the two highest doses are very effective in eliciting an arrestment response by bed bugs (Table 7). To determine the effect of dimethylaminoethanol or histamine in the 2-component blend, Experiments 35-37 tested dimethylaminoethanol and histamine alone at 200 μg each and in binary combination at a ratio of 20 μg:200 μg. The data reveal that histamine, but not dimethylaminoethanol, is

bioactive on its own, and at the dose tested is as effective as the 2-component blend in arresting bed bugs.
[0118] Experiments 23-37 do not define a clear role for dimethylaminoethanol. On the one hand, when dimethylaminoethanol and histamine were offered at a ratio of 2 μg:20 μg in Experiment 32, there was no preferential selection of the baited chamber. However, when the dose of dimethylaminoethanol was raised to 20 μg, equal to that of histamine, in Experiments 24 and 29, a preference for the baited chamber appeared. On the other hand, unlike histamine in Experiment 37, dimethaminoethanol alone in Experiment 36 was inactive. Moreover, increasing the dose of dimethylaminoethanol 10-fold in Experiments 32, while the dose of histamine was held constant, did not result in an increased response.
[0119] Table 7 illustrates the responses in the three-dish, dual-choice olfactometer of bed bug nymphs in Experiments 23-37 to 1- or 2-component baits of dimethylaminoethanol (D) and histamine (H); numbers in parentheses indicate amounts in micrograms.
TABLE-US-00007 No. No. No. insects insects insects Exp. in bait in control not no. Bait* chamber chamber responding 23 D (2):H (2) 4 4 4 24 D (20):H (20) 8 2 2 25 D (200):H (200) 12 0 0 26 D (20):H (20) (base) 10 1 1 27 D (20):H (20) (salt) 5 4 3 28 D (20):H (20) 9 1 2 (base/salt) 29 D (20):H (20) 15 5 4 30 D (200):H (20) 13 4 7 31 D (20):H (200) 20 2 2 32 D (2):H (20) 6 5 2 33 D (20):H (200) 10 2 0 34 D (200):H (2000) 10 1 1 35 D (20):H (200) 33 11 4 36 D (200):H (0) 14 15 19 37 D (0):H (200) 35 10 3
Example 9
Effect of Additional Components (L-valine, L-alanine, N-acetylglucosamine, 3-hydroxy-kynurenine O-sulfate) on the Blend of Histamine and Dimethylaminoethanol
[0120] With other constituents being present in methanol extracts of bed bug exuviae or feces (Table 5), Experiments 38 and 39, and 40 and 41, tested whether the 2-component blend of dimethylaminoethanol and histamine (20 μg: 200 μg) would become more effective through the addition of L-valine, L-alanine and N-acetylglucosamine (Experiment 39), or the addition of 3-hydroxy-kynurenine O-sulfate (Experiment 41). The data reveal that the effectiveness in arresting bed bugs of the 2-component blend of dimethylaminoethanol and histamine (Experiments 38 and 40) could not be improved by the addition of other components (Table 8).
[0121] Table 8 illustrates the responses of bed bug nymphs in the three-dish, dual-choice olfactometer in Experiments 38-42 to the 2-component bait of dimethylaminoethanol (D) and histamine (H) or the same bait with additional components [L-valine (V), L-alanine (A), N-acetylglucosamine (N-ac), 3-hydroxykynurenine O-sulfate (3-Hyd)] identified in methanol extracts of bed bug exuviae and feces; numbers in parentheses indicate amounts in micrograms.
TABLE-US-00008 No. No. No. insects insects insects Exp. in bait in control in not no. Bait* chamber chamber responding 38 D (20):H (200) 13 3 2 39 D (20):H 12 4 2 (200):V (20):A (20):N-ac (20) 40 D (20):H (200) 10 2 0 41 D (20):H 9 3 0 (200):3-Hyd (200)
Example 10
Comparative Responses of Bed Bug Nymphs, Adult Males and Adult Females to the Blend of Dimethylaminoethanol and Histamine
[0122] To determine whether the 2-component blend of dimethylaminoethanol and histamine elicits arrestment response of immature and mature stages of bed bugs, Experiments 42-44 tested dimethylaminoethanol and histamine (20 μg: 200 μg) for the responses of bed bug nymphs, adult males and adult females. The data reveal that the 2-component blend is equally effective in inducing arrestment responses of nymphs, and adult males and females (Table 9).
[0123] Table 9 illustrates the responses in the three-dish, dual-choice olfactometer of bed bug nymphs, adult males or adult females to the 2-component blend of dimethylaminoethanol (D) and histamine (H); numbers in parentheses indicate amounts in micrograms.
TABLE-US-00009 No. No. No. insects insects insects Exp. Insects in bait in control not no. Bait* tested chamber chamber responding 42 D (20):H (200) nymphs 18 4 2 43 D (20):H (200) males 19

1 4 44 D (20):H (200) females 19 2 3

Example 11

Identification of Candidate Volatile Pheromone Components in Bed Bug Feces

[0124] With evidence that dimethylaminoethanol has only a limited pheromonal role (Table 7), and with histamine being less volatile and thus likely serving as an arrestant rather than an attractant, the search was continued for attractive volatile pheromone components. The focus was also shifted to bed bug feces which are present in natural bed bug shelters along with exuviae.

[0125] Pieces of filter paper (5×10 cm) exposed to the feces of approximately 300 bed bugs over a period of four weeks were cut into small sections of 0.75×0.5 cm, each of which were analyzed with an Agilent Headspace Analyzer coupled to a Varian 2000 Ion Trap GC-MS fitted with a DB-5 MS GC column (30 m×0.25 μm ID). After placing the feces-stained paper into a 20-ml vial, it was sealed with a crimped cap with a 20-mm OD white silicon septum and heated to 90° C. for 5 min. The airborne headspace volatiles were withdrawn with an automated syringe and subjected to coupled gas GC-MS analysis, using the following temperature program: 50° C. for 5 min, then 10° C. per min to 280° C. The analysis revealed a complex blend of 15 oxygen- or sulphur-containing volatile components, consisting of six aldehydes (butanal, pentanal, hexanal, (E)-2-hexenal, (E)-2-octenal, benzaldehyde), one alcohol (benzyl alcohol), three ketones (2-hexanone, acetophenone, verbenone), three esters (methyl octanoate, ethyl octanoate, pentyl hexanoate) and two sulfides (dimethyl disulfide, dimethyl trisulfide). All of these 15 compounds were considered candidate pheromone components to be tested for attraction of bed bugs in dual-choice olfactometer bioassays (see EXAMPLE 2; FIG. **1**).

Example 12

Responses of Adult Male Bed Bugs to Synthetic Blends at Three Doses of Bed Bug Feces Volatiles

[0126] To determine whether a blend of the 15 bed bug feces volatiles that were identified in bed bug feces (see EXAMPLE 11) attracts bed bugs and thus contains one or more bed bug pheromone components, the synthetic blend (SB) was tested at a medium dose (50 μg; Experiment 45), a low dose (5 μg; Experiment 46), and a very low dose (0.5 μg; Experiment 47) for the responses of bed bugs in three-dish, dual-choice olfactometers (see EXAMPLE 2, FIG. **19**). At each dose tested, all components were formulated in equal amounts in mineral oil which was pipetted into the inverted lid of a 4-ml vial. The vial was then placed on top of the corrugated cardboard shelter (see FIG. **1**) in the randomly assigned treatment dish of the olfactometer (see EXAMPLE 2). The control stimulus consisted of the same type of lid on the corrugated cardboard of the control dish of the olfactometer containing mineral oil without synthetic test chemicals.

[0127] In Experiments 45-47, synthetic blends of bed bug feces volatiles at a medium, low and very low dose, all attracted more bed bug males than did control stimuli, indicating that the blend contained one or more bed bug pheromone components. The medium dose was selected for use in further experiments.

[0128] Table 10 illustrates the responses of bed bug males in the three-dish, dual-choice olfactometer to a synthetic blend (SB) of 15 bed bug feces volatiles (EXAMPLE 11) tested at a medium dose (50 μg), low dose (5 μg), and very low dose (0.5 μg); at each dose, all components were formulated in equal amounts in mineral oil.

TABLE-US-00010

Exp. no.	Bait*	No. insects in bait chamber	No. insects in control chamber	No. insects not responding
45	SB (50 μg)	29	7	0
46	SB (5.0 μg)	21	12	3
47	SB (0.5 μg)	24	10	2

Example 13

Determination of Pheromone Components in Synthetic Blend of Bed Bug Feces Volatiles

[0129] To determine the important component(s) in the 15-component synthetic blend (SB) of bed bug feces volatiles that attract bed bugs (EXAMPLE 12; Table 10), Experiments 48-52 compared their responses to the complete SB of all 15 components (Exp. 48), with those to SBs that lacked groups of related organic chemicals, i.e., esters (methyl octanoate, ethyl octanoate, pentyl hexanoate; Exp. 49), aldehydes (butanal, pentanal, hexanal, (E)-2-hexenal, (E)-2-octenal, benzyl aldehyde; Exp. 50), sulfides (dimethyl disulfide, dimethyl trisulfide; Exp. 51), and ketones/alcohol (2-hexanone, acetophenone, verbenone, benzyl alcohol; Exp. 52). All SBs were formulated and bioassayed as described in EXAMPLE 11.

[0130] In Experiment 48 (Table 11), the 15-component SB strongly attracted adult male bed bugs, confirming that this SB contained one or more bed bug pheromone components. Similarly, in Experiment 49 (Table 11), the SB lacking esters strongly attracted adult male bed bugs, indicating that esters are not an important part of the bed bug pheromone blend. Conversely, shelters baited with SBs lacking aldehydes (Exp. 50), sulfides (Exp. 51), or ketones and alcohol (Exp. 52), all failed to capture significantly more adult male bed bugs than control shelters, indicating that one or more of the aldehydes, sulfides, and ketones or alcohol are important bed bug pheromone components.

[0131] Table 11 illustrates the responses of adult male bed bugs in the three-dish, dual-choice olfactometer to: a 15-component synthetic blend (SB) comprising 6 aldehydes (butanal, pentanal, hexanal, (E)-2-hexenal, (E)-2-octenal, benzyl aldehyde), one alcohol (benzyl alcohol), 3 ketones (2-hexanone, acetophenone, verbenone), 3 esters (ethyl octanoate, methyl octanoate, pentyl hexanoate), and 2 sulfides (dimethyl disulfide and dimethyl trisulfide) (Exp. 48, 53 and 57); said 15-component blend minus said 3 esters, 6 aldehydes, 2 sulfides, or one alcohol and 3 ketones (Exp. 49-52, respectively); said 15-component blend minus benzaldehyde and benzyl alcohol (Exp. 54), verbenone (Exp. 55), or (E)-2-hexenal and (E)-2-octanal (Exp. 56); a 6-component synthetic blend (6-Comp. SB) comprising (E)-2-hexenal, (E)-2-octenal, 2-hexanone, acetophenone, dimethyl disulfide and dimethyl trisulfide (Exp. 58, 59, 62 and 65); and said 6-component synthetic blend minus acetophenone (Exp. 60), 2-hexanone (Exp. 61), dimethyl disulfide (Exp. 63), dimethyl trisulfide (Exp. 64), (E)-2-hexenal (Exp. 66), or (E)-2-octenal (Exp. 67). Experiments in each of the following groups were run concurrently: 48-52, 53-56, 57-58, 59-61, 62-64, and 65-67. All blends were tested at a dose of 50 μg.

TABLE-US-00011 No. No. No. insects insects insects Exp. in in control in not no. Bait bait chamber chamber responding 48 SB 18 4 2 49 SB minus esters 16 4 4 50 SB minus aldehydes 11 7 6 51 SB minus sulfides 10 7 7 52 SB minus ketones and 11 10 3 alcohol 53 SB 12 6 2 54 SB minus benzaldehyde 13 6 1 and benzyl alcohol 55 SB minus verbenone 14 6 0 56 SB minus 9 9 2 (E)-2-hexenal and (E)-2-octenal 57 SB 15 5 0 58 6-Comp. SB 14 2 1 59 6-Comp. SB 20 3 1 60 6-Comp. SB minus 20 3 1 acetophenone 61 6-Comp. SB minus 13 9 1 2-hexanone 62 6-Comp. SB 20 3 1 63 6-Comp. SB minus 18 6 0 dimethyl trisulfide 64 6-Comp. SB minus 17 6 1 dimethyl disulfide 65 6-Comp. SB 19 5 0 66 6-Comp. SB minus 18 5 1 (E)-2-hexenal 67 6-Comp. SB minus 15 8 1 (E)-2-hexenal

[0132] To narrow down the specific aldehyde(s), ketone(s) or sulfide(s) that constitute bed bug pheromone components, follow-up and concurrently-run experiments again tested the complete SB (Exp. 53), and SBs lacking both benzyl aldehyde and benzyl alcohol (Exp. 54), verbenone (Exp. 55), or both (E)-2-hexenal and (E)-2-octenal (Exp. 56).

[0133] In Experiment 53 (Table 11), shelters baited with the 15-component SB captured twice as many adult bed bug males as did control shelters. Similar data were obtained with SBs lacking both benzyl aldehyde and benzyl alcohol (Exp. 54) or lacking verbenone (Exp. 55), indicating that none of these three compounds is an important bed bug pheromone component. Conversely, shelters baited with the SB lacking both (E)-2-hexenal and (E)-2-octenal (Exp. 56) failed to capture more adult male bed bugs that did control shelters, indicating that one or both of these two aldehydes are bed bug pheromone components.

[0134] The combined data of Experiments 48-56 indicated that the major attractive pheromone components of bed bugs are among the following six compounds: (E)-2-hexenal, (E)-2-octenal, dimethyl disulfide, dimethyl trisulfide, acetophenone and 2-hexanone. To ascertain whether this 6-component synthetic blend (6-Comp. SB) was as effective as the 15-comp. SB in attracting bed bugs, Experiments 57 and 58 tested SB and 6-Comp. SB versus control stimuli.

[0135] In Experiment 57 (Table 11), the 15-Comp. SB, as expected, attracted significantly more adult male bed bugs than did the control stimulus. In Experiment 58, the 6-Comp. SB attracted significantly more adult male bed bugs than did the control stimulus. Because SB and 6-Comp. SB appeared equally capable of attracting bed bugs, it was concluded that all major pheromone components are present in the 6-Comp. SB.

[0136] To determine the important ketone(s) in the 6-Comp. SB, follow-up and concurrently-run experiments compared responses to the 6-Comp. SB (Exp. 59), the 6-Comp. SB lacking acetophenone (Exp. 60), and the 6-Comp. SB lacking 2-hexanone (Exp. 61).

[0137] In Experiment 59 (Table 11), the 6-Comp. SB attracted significantly more adult male bed bugs than did the control stimulus. Similarly, in Experiment 60 the 6-Comp. SB lacking acetophenone attracted significantly more adult male bed bugs than did the control stimulus, indicating that acetophenone is not an important bed bug pheromone component. Conversely, in Experiment 61, the 6-Comp. SB lacking 2-hexanone failed to attract significant numbers of adult male bed bugs, indicating that 2-hexanone is a key bed bug pheromone component.

[0138] To determine the key sulfide(s) in the 6-Comp. SB, the next three follow-up and concurrently-run experiments compared responses to the 6-Comp. SB (Exp. 62), the 6-Comp. SB lacking dimethyl disulfide (Exp. 63), and the 6-Comp. SB lacking dimethyl trisulfide (Exp. 64).

[0139] In Experiment 62, the 6-Comp. SB attracted significantly more adult male bed bugs than did the control stimulus. Similarly, the two 6-Comp. SBs lacking either dimethyl disulfide (Exp. 63) or dimethyl trisulfide (Exp. 64) were still more effective than control stimuli in attracting adult male bed bugs. However, the 6-Comp. SB with both of these sulfides (Exp. 62) appeared relatively more attractive than SBs containing just one sulfide (Exp. 63 and 64), suggesting that both of these sulfides are important pheromone components and should be included in operational lures.

[0140] To determine the key aldehyde(s) of the 6-Comp. SB, the next three follow-up and concurrently run experiments compared responses to the 6-Comp. SB (Experiment 65), the 6-Comp. SB lacking (E)-2-hexenal (Exp. 66), and the 6-Comp. SB lacking (E)-2-octenal (Exp. 67).

[0141] In Experiment 65 (Table 11), the 6-Comp. SB attracted significantly more adult male bed bugs than did the control stimulus. Similarly, in Experiment 66 the 6-Comp. SB lacking (E)-2-hexenal attracted significantly more adult male bed bugs than did the control stimulus. Conversely, in Experiment 67 the 6-Comp. SB lacking (E)-2-octenal attracted barely twice as many adult male bed bugs than did the control stimulus. The data in combination reveal that (E)-2-octenal is a relatively more important pheromone component than (E)-2-hexenal. Nonetheless, both aldehydes are to be included in a commercial bed bug pheromone lure for optimal attractiveness.

Example 14

Responses of Bed Bugs to Synthetic Pheromone Lures in Large Bioassay Arenas

[0142] Experiments 68-75 were carried out to ascertain whether bed bugs respond to synthetic pheromone not only in small olfactometers (EXAMPLE 2; FIG. **1**; Tables 7-11) but also in large bioassay arenas. Experiments 68-75 also investigated the effect of the volatile pheromone components (VPCs) (E)-2-hexenal, (E)-2-octenal, dimethyl disulfide, dimethyl trisulfide and 2-hexanone, and the effect of the less-volatile pheromone component histamine (H), on attraction and arrestment of bed bugs. Treatment stimuli consisted of filter paper (4×2.2 cm) impregnated with H (2,000 μg) and covered with a piece of corrugated cardboard shelter (3×2.2 cm), and of VPCs formulated at a high dose (500 μg; Experiments 68-74) or a medium dose (50 μg; Exp. 75) in mineral oil (0.5 mL) and pipetted into the inverted lid of a 20-ml scintillation vial resting on top of the shelter (FIG. **19**). Control stimuli were of identical design but the filter paper contained no H, and the mineral oil contained no VPCs.

[0143] Specifically, Experiment 68 tested the complete synthetic pheromone blend consisting of VPCs and H versus a control stimulus, whereas concurrently-run Experiment 69 tested a partial pheromone blend consisting of only VPCs versus a control stimulus.

[0144] In Experiment 68 (Table 12), 15 of the 16 male bed bugs that were tested individually

responded to the complete pheromone blend, indicating that the complete blend contained all important bed bug pheromone components. In Experiment 69, in contrast, only 6 of the 16 male bed bugs tested responded to the partial pheromone blend consisting of VPCs without histamine, confirming that histamine is an important bed bug pheromone component (see also Tables 7-9), and that H serves the role of arresting bed bugs at a shelter once they have been attracted to it by the VPCs.
[0145] Table 12 illustrates the responses of adult male bed bugs to complete or partial synthetic pheromone blends comprising the volatile pheromone components (VPCs) (E)-2-hexenal, (E)-2-octenal, dimethyl disulfide, dimethyl trisulfide and 2-hexanone, and/or the less-volatile pheromone component histamine (H). VPCs in experiments 68-74 were tested at 500 μg and in experiment 75 at 50 μg. Histamine was tested at 2,000 μg. Experiments 68-69, 70-71, 72-73, and 74-75 were run concurrently.
TABLE-US-00012 No. No. No. insects Exp. insects in insects in not no. Shelter 1 Shelter 2 shelter 1 shelter 2 responding 68 VPC + H unbaited 15 0 1 69 VPC unbaited 6 3 7 70 H unbaited 12 1 5 71 VPC + H unbaited 17 1 3 72 VPC H 2 14 4 73 VPC + H unbaited 15 1 4 74 VPC + H H 17 5 3 75 VPC + H H 10 0 6
[0146] To determine the effect of VPCs in the pheromone blend, we then tested concurrently histamine alone versus a blank control (Exp. 70), and the complete pheromone blend of VPCs and histamine versus a blank control (Exp. 71).
[0147] In Experiment 70 (Table 12), 12 bed bugs responded to the shelter with histamine-impregnated filter paper, whereas only one bed bug responded to the shelter with blank filter paper, indicating again that bed bugs are arrested in the presence of histamine. In Experiment 71, 17 bed bugs responded to the shelter associated with the complete pheromone blend of VPCs and histamine, and only one bed bug responded to the control shelter, suggesting that the complete pheromone blend was possibly more effective than the partial blend (tested in Exp. 70) in attracting or arresting bed bugs.
[0148] To compare further the relative importance of histamine and VPCs as bed bug pheromone components, we tested concurrently the responses of bed bugs to shelters baited with histamine or VPCs (Exp. 72), and the complete pheromone blend of VPCs and histamine versus a blank control (Exp. 73).
[0149] In Experiment 72 (Table 12), 14 bed bugs responded to shelter baited with histamine and two bed bugs responded to shelter baited with VPCs, indicating that histamine had a stronger effect on the bed bugs' decision which shelter to select. In Experiment 73, 15 bed bugs responded to shelter baited with the complete pheromone blend (VPCs+histamine) and only one bed bug responded to unbaited control shelter, confirming the superior effect of the complete bed bug pheromone blend.
[0150] With histamine strongly arresting bed bugs at a shelter (see Exp. 70 & 72 in Table 12), we then explored whether the effect of histamine could be enhanced by adding VPCs to histamine. Accordingly, we tested the responses of bed bugs to histamine or to histamine plus VPCs, both at high dose of VPCs (500 μg; Exp. 75) and at a medium dose of VPCs (50 μg; Exp. 74).
[0151] In Experiments 74 and 75, the complete pheromone blend of histamine plus VPCs at a high dose and at a medium dose attracted and arrested significantly more bed bugs than did histamine alone, clearly revealing a synergistic effect between VPCs (attracting bed bugs) and histamine (arresting bed bugs).
Example 15
Responses of Bed Bugs to Synthetic Pheromone Lures in a Heavily Infested Residential Apartment
[0152] To determine whether bed bugs responded to synthetic pheromone lures not only in large arena laboratory bioassays but also in infested premises (residential apartments), we placed pheromone-baited shelter traps (see FIG. **19**) in apartments with known or suspected bed bug infestations. The shelter was identical to that described in EXAMPLE 13 except that the histamine-

impregnated filter paper or control filter paper was glued to the corrugated cardboard shelter for ease of shelter placements. Based on 24-hour trapping results in multiple apartments, we selected a single, heavily infested apartment (hereafter referred to as Room 106) for further testing of synthetic bed bug pheromone lures.

[0153] To obtain field evidence for synergistic interaction between the VPCs and histamine in synthetic pheromone lures (see Exp. 74 & 75 in Table 12), two consecutive sets of two experiments were run (Set 1: Exp. 76 and 77; Set 2: Exp. 78 and 79) in Room 106. Replicates (n=15-20) of each experiment consisted of paired shelters placed against a wall or furniture, with approximately 30-cm spacing between paired shelters and >1 m spacing between shelter pairs. For each shelter pair within each experiment, pheromone and control treatments were randomly assigned, and placement of replicates was alternated between the two experiments of Set 1 and Set 2. Because not more than 10 shelter pairs could be accommodated in Room 106 at any time, experimental data were collected over several consecutive 24-h periods. Every morning at 9:00, shelters were collected from Room 106 and immediately placed into separate labelled zipper lock bags which were then put on dry ice to kill the bed bugs that had entered a shelter. Bed bugs in each shelter were later counted in the laboratory under a microscope, noting their developmental stage (1.sup.st, 2.sup.nd, 3.sup.rd, or 4.sup.th instar, adult), sex (male, female), and evidence (or not) for blood feeding prior to entry into a shelter. Immediately following the removal of bed bug shelters from Room 106 each morning, new pre-prepared shelters were pheromone-baited (treatment) or not (control) and their position within shelter pairs randomly assigned (see above), again alternating placement of shelter pairs between Experiments 76 and 77 (Set 1) and between Experiments 78 and 79 (Set 2).

[0154] To determine the effectiveness of synthetic pheromone lures in the presence or absence of the volatile pheromone components (VPCs) (E)-2-hexenal, (E)-2-octenal, dimethyl disulfide, dimethyl trisulfide, 2-hexanone (that we predicted would attract bed bugs to a shelter), and the less volatile component histamine (that we predicted would retain bed bugs at a shelter), we compared the responses of bed bugs to complete and partial synthetic pheromone lures in Experiments 76 and 77, and in Experiments 78 and 79, with experiments in each set run concurrently.

[0155] In Experiment 76 we tested shelters baited with the complete synthetic pheromone lure (VPCs+histamine) versus unbaited shelters, and in Experiment 77 we tested shelters baited with VPCs only versus unbaited shelters. Analogously, in Experiment 78 we tested shelters baited with the complete synthetic pheromone lure (VPCs+histamine) versus unbaited shelters, and in Experiment 79 we tested shelters baited with histamine only versus unbaited shelters.

[0156] In Experiment 76 (Table 13), shelters baited with the complete synthetic pheromone blend attracted and retained, on average, 21.4 bed bugs compared to 4.8 bed bugs, on average, in control shelters. These data indicate that the complete synthetic pheromone lure had a significant effect on attracting and retaining bed bugs. Conversely, in Experiment 77 shelters baited with only the VPCs attracted and retained, on average, only 10.3 bed bugs compared to 4.7 bed bugs, on average, in control shelters.

[0157] Table 13 illustrates the responses of nymphal and adult bed bugs in a residential apartment to corrugated cardboard shelter traps (see FIG. **19**) baited with the complete synthetic pheromone blend consisting of the volatile pheromone components (VPCs) (E)-2-hexenal, (E)-2-octenal, dimethyl disulfide, dimethyl trisulfide and 2-hexanone (total amount 500 μg) and of the less volatile pheromone component histamine (H) (2,000 μg), or baited with partial pheromone blends lacking either the VPCs or histamine. The pheromone-baited shelter and unbaited control shelter in each experimental replicate were separated by at least 30 cm, with ≥1-m spacing between replicates. There was alternate placement of replicates of concurrently run Experiments 76 and 77 (20 replicates each), and of concurrently run Experiments 78 and 79 (15 replicates each).

TABLE-US-00013 Mean ± SE number Mean (±SE) Exp. insects in number insects no. Bait* baited shelter in control shelter 76 VPCs + H 21.4 ± 5.93 4.8 ± 1.82 77 VPCs 10.3 ± 3.45 4.7 ± 1.73 78 VPCs + H 24.86 ± 6.84 3.73 ± 1.43 79 H 6.06 ± 1.28 4.53 ± 1.8

[0158] The combined data from Experiments 76 and 77 reveal that the complete synthetic pheromone lure is superior to a partial pheromone lure (lacking histamine) in attracting and retaining bed bugs.
[0159] In Experiment 78 (Table 13), shelters baited with the complete synthetic pheromone blend attracted and retained, on average, 24.86 bed bugs compared to 3.73 bed bugs, on average, in control shelters. These data indicate again that the complete synthetic pheromone lure had a significant effect on attracting and retaining bed bugs. Conversely, in Experiment 79 shelters baited with a partial pheromone blend containing only histamine attracted and retained, on average, only 6.06 bed bugs compared to 4.53 bed bugs, on average, in control shelters.
[0160] The combined data from Experiments 76-79 reveal that the complete synthetic pheromone blend is most effective in attracting and retaining bed bugs. The superior performance of the complete synthetic pheromone blend is due to the combined effects of two types of pheromone components, the volatile components (VPCs) that attract bed bugs to a shelter, and the less-volatile pheromone component histamine that arrests bed bugs in a shelter after they have been attracted to it.
[0161] Pheromone-baited shelters contained all nymphal instars (fed and non-fed) as well as adult males and females (fed and non-fed), clearly indicating that the complete synthetic pheromone attracts and retains bed bugs irrespective of their developmental stage, gender or physiological condition.

Example 16

Responses of Bed Bugs to Synthetic Pheromone Lures in Residential Apartments with Light Bed Bug Infestations

[0162] To ascertain whether bed bugs responded to the complete synthetic pheromone not only in residential apartments heavily infested with bed bugs (see EXAMPLE 15; Table 13) but also in apartments with very light bed bug infestations, trapping was done in eight residential apartments in three separate buildings where tenants suspected they had bed bugs according to reports of a pest management professional. In each of these apartments, bedding, mattresses and other furniture were carefully inspected for the presence of live bed bugs and for evidence of bed bug activity such as fecal spots on bed sheets. In Experiment 80, 1-5 pairs of bed bug shelter traps (see FIG. **19**) were placed in each apartment for a total of 20 pairs. One shelter in each pair was pheromone-baited [VPCs: (E)-2-hexenal, (E)-2-octenal, dimethyl disulfide, dimethyl trisulfide, 2-hexanone (500 μg) plus histamine (2,000 μg)]; the other was left unbaited. Shelter placement and spacing within and between shelter pairs proceeded as described in EXAMPLE 15. Shelters were retrieved 24 h after the placement, immediately placed into labelled separate zipper lock bags and put on dry ice to kill all captured bed bugs.
[0163] In Experiment 80, 26 of 27 bed bugs captured were present in pheromone-baited shelters; only one bed bug was present in an unbaited control shelter. Shelter traps in all rooms in which at least one live bed bug was seen at the start of the experiment captured at least one bed bug.
[0164] These data indicate that the 6-component synthetic pheromone lure comprised of (E)-2-hexenal, (E)-2-octenal, dimethyl disulfide, dimethyl trisulfide, 2-hexanone and histamine is capable of attracting and retaining bed bugs in lightly-infested residential apartments and that this novel pheromone lure has the potential to become an effective tool for detecting bed bug infestations in residential and commercial premises.

Example 17

Evaluation of a Bed Bug Trap Against Adult Bed Bugs

[0165] Experiment 81 utilized bioassay arenas comprised of 19-liter plastic containers with paper toweling taped on the inside bottom to afford traction for walking bed bugs. Bed bugs were purchased from North Carolina State University. Prior to the initiation of testing, insects were observed for 72 hours to ensure robust health. Each arena received 15 adult bed bugs per 55-minute replicate and either an unbaited control bed bug trap according to an embodiment (e.g., as shown in

FIGS. **1-15**), a baited bed bug trap according to an embodiment (e.g., as shown in FIGS. **1-15**), or a Hot Shot® Bed Bug Glue Trap (Spectrum Brands, Middleton, WI). The baited bed bug trap is baited with an attractant (**150**, **152**, **154**) that, when activated, releases the five VPCs listed in Example 15 [(E)-2-hexenal, (E)-2-octenal, dimethyl disulfide, dimethyl trisulfide, 2-hexanone]. Each treatment was replicated five times. Data were analyzed by ANOVA followed by Tukey's HSD test, $\alpha=0.05$.
[0166] The baited bed bug trap captured 4.2 bed bugs in 55 minutes; this catch was 2.6 times that in the same trap without an attractant lure and 10.5 times that in the Hot Shot® Bed Bug Glue Trap (Table 14). The mean catch in the baited bed bug traps was significantly greater than the mean catch in either of the other two traps, which were not different from each other. These data demonstrate that, when baited with a volatile attractant composition, the bed bug trap according to an embodiment has the potential to detect a bed bug infestation within one hour.
[0167] Table 14. Results of Experiment 81 (N=5 replicates) showing catches of bed bugs in arena olfactometers (15 adult bed bugs released per replicate) in 55-minute periods in attractant-baited and unbaited bed bug traps compared with the catch in a widely-available commercial trap.
TABLE-US-00014 TABLE 14 Treatment Mean catch in 55 minutes (±SE)* Unbaited bed bug trap 1.60 ± 0.87 b Baited bed bug trap 4.20 ± 0.37 a Hot Shot ® Bed Bug Glue Trap 0.40 ± 0.24 b *F2, 12 = 11.7917, P = 0.0015. Means followed by the same letter are not significantly different, Tukey's HSD test, P < 0.05.
Unbaited Bed Bug Trap 1.60±0.87 b
[0168] This application is intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains and which fall within the limits of the appended claims. Accordingly, the scope of the claims should not be limited by the preferred embodiments set forth in the description, but should be given the broadest interpretation consistent with the description as a whole.

Claims

1. An insect trap comprising: a first housing; a reservoir at least partially defined by the first housing; a protrusion disposed within the reservoir and extending inward from an interior surface of the first housing, the protrusion configured to pierce a package disposed within the reservoir; an actuator connected to the protrusion such that when the actuator is moved, the protrusion moves with the actuator within the reservoir; a second housing moveably engaging the first housing; a trap chamber partially formed by the first and second housings; an inlet into the trap chamber; and wherein the second housing is movable between a first position where the inlet is open and a second position where the inlet is closed.
2. (canceled)
3. The insect trap of claim 1, wherein the trap chamber comprises a floor, further comprising an adhesive coated on a surface of the floor.
4. The insect trap of claim 3, wherein the first housing comprises a bottom wall and wherein the floor is adhered to the bottom wall using at least a portion of the adhesive.
5. (canceled)
6. The insect trap of claim 1, wherein the trap chamber comprises a floor, and wherein the first housing comprises a bottom wall and wherein the bottom wall forms the floor.
7. The insect trap of claim 6, further comprising an adhesive at least partially coating a surface of the floor.
8. The insect trap of claim 1, wherein the trap chamber comprises a floor, wherein the first housing comprises a bottom wall, and wherein at least a portion of the bottom wall adjacent to the floor is at an elevation above an elevation of the floor, forming a pit to trap insects therein.

9. The insect trap of claim 8, where the bottom wall rises in elevation from the inlet to a point in the trap chamber adjacent the floor.
10. (canceled)
11. (canceled)
12. The insect trap of claim 1, further comprising an attractant configured to attract insects and to be disposed within the reservoir, wherein the attractant comprises an attractant composition configured to attract insects, a first package enclosing the attractant composition, and a second package enclosing the first package therein and configured to be impermeable to gas.
13-16. (canceled)
17. The insect trap of claim 1, wherein the trap chamber comprises a floor, and wherein the first housing comprises: a first portion, the reservoir being disposed within the first portion of the first housing; a second portion spaced-apart from the first portion of the first housing; and a space defined between the first and second portions of the first housing; wherein the second housing is positioned over the space and movingly engaged with the first and second portions of the first housing to form the trap chamber and permit the second housing to move between the first and second positions.
18. The insect trap of claim 17, wherein a first tab extends from the first portion of the first housing into the space and a second tab extends into the space from the second portion of the first housing, and wherein the second housing includes a first detent configured to receive the first tab and a second detent configured to receive the second tab.
19. The insect trap of claim 18, wherein when the first and second tabs are engaged with the first and second detents, the second housing is held in the first position.
20. The insect trap of claim 19, wherein when the second housing is moved from the first position to the second position, the second housing is moved below the first and second tabs such that the first and second tabs act as a stop to hold the second housing in the second position.
21. The insect trap of claim 17, wherein a longitudinal chamber member connects the first portion of the first housing to the second portion of the first housing.
22. The insect trap of claim 21, wherein a plurality of interior channel walls extend perpendicularly from the longitudinal chamber member, forming a plurality of channels between each set of adjacent interior channel walls, each channel connected to the inlet when the second housing is in the first position.
23. The insect trap of claim 17, wherein the second housing slideably engages the first and second housing portions, enabling the second housing to slide between the first and second positions.
24. The insect trap of claim 1, wherein the first housing comprises a plurality of interior channel walls disposed within the trap chamber forming a plurality of sub-chambers, each sub-chamber connected to the inlet when the second housing is in the first position.
25-30. (canceled)
31. The insect trap of claim 1, wherein the first housing comprises a bottom wall disposed at the inlet that is configured to provide grip for insects crawling into the trap chamber via the inlet.
32-33. (canceled)
34. The insect trap of claim 1, wherein the trap chamber comprises a retention mechanism selected from the group consisting of histamine, a chemical toxin, diatomaceous earth, amorphous silica, and biological control agents.
35-43. (canceled)
44. An insect trap comprising: a housing; a reservoir at least partially defined by the housing; a protrusion extending from an interior surface of the housing; an actuator disposed upon the housing and configured to move the protrusion within the reservoir; a trap chamber at least partially formed by the housings and constructed to capture insects; an outer package disposed within the reservoir and impermeable to gas; an inner package enclosed within the outer package, the inner package being permeable to gas; and an attractant composition disposed within the inner package; wherein

the inner package is configured to release the attractant composition at a release rate from about 15 μg per day to about 400 μg per day.

45-47. (canceled)

48. An insect trap comprising: a housing; a reservoir at least partially defined by the first housing; a protrusion disposed within the reservoir and extending inward from an interior surface of the first housing, the protrusion configured to pierce a package disposed within the reservoir; an actuator connected to the protrusion such that when the actuator is moved, the protrusion moves with the actuator within the reservoir; a substrate defining a plurality of channels; a trap chamber partially formed by the housing and the substrate; an inlet into the trap chamber.

49-64. (canceled)